MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC ASSESSMENT – 11** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions

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Q1. As the acute angle θ increases from 0° to 90°, the value of $\sin\theta$: [1]

(A) Increases from 0 to 1 (B) Decreases from 1 to 0

(C) Remains constant (D) Decreases then increases

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Increases from 0 to 1

Justification: Standard values: $\sin 0^\circ = 0$, $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{1}{\sqrt{2}}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\sin 90^\circ = 1$.

Q2. If $\tan\theta = \frac{4}{3}$, then the value of $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}}$ is: [1]

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{1}{2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\frac{1}{3}$

Justification: $\tan\theta = \frac{4}{3}$ gives a 3–4–5 right triangle with $\sin\theta = \frac{4}{5}$. Thus $\sqrt{\frac{1-4/5}{1+4/5}} = \sqrt{\frac{1/5}{9/5}} = \sqrt{\frac{1}{9}} = \frac{1}{3}$.

Q3. The value of $(\sec^2\theta - 1)(\csc^2\theta - 1)$ is equal to: [1]

(A) 0 (B) 1 (C) -1 (D) 2

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 1

Justification: Using $\sec^2\theta - 1 = \tan^2\theta$ and $\csc^2\theta - 1 = \cot^2\theta$, their product is $\tan^2\theta \cdot \cot^2\theta = 1$.

Q4. If $\cos\theta = \frac{1}{2}$ and $\sin\phi = \frac{1}{2}$ for acute angles θ and ϕ, then $\theta + \phi$ is: [1]

(A) 30° (B) 60° (C) 90° (D) 120°

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (C) 90°

Justification: $\cos\theta = \frac{1}{2} \implies \theta = 60^\circ$, and $\sin\phi = \frac{1}{2} \implies \phi = 30^\circ$. Hence $\theta + \phi = 60^\circ + 30^\circ = 90^\circ$.

Q5. If $a\cos\theta + b\sin\theta = 4$ and $a\sin\theta - b\cos\theta = 3$, then $a^2 + b^2$ is equal to: [1]

(A) 7 (B) 12 (C) 25 (D) 5

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (C) 25

Justification: Squaring and adding gives $a^2(\cos^2\theta + \sin^2\theta) + b^2(\sin^2\theta + \cos^2\theta) = 4^2 + 3^2 \implies a^2 + b^2 = 16 + 9 = 25$.

Q6. Assertion (A): The value of $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$ is 1.

Reason (R): $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\sin 30^\circ = \frac{1}{2}$, $\cos 60^\circ = \frac{1}{2}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Justification: Substituting from (R): $\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{3}{4} + \frac{1}{4} = 1$. Both are true and (R) explains (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. If $3\tan\theta = 4$, evaluate $\dfrac{3\sin\theta + 2\cos\theta}{3\sin\theta - 2\cos\theta}$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Divide Numerator & Denominator by $\cos\theta$ [1 Mark]

$$\frac{3\sin\theta + 2\cos\theta}{3\sin\theta - 2\cos\theta} = \frac{3\tan\theta + 2}{3\tan\theta - 2}$$

Step 2: Substitute $3\tan\theta = 4$ [1 Mark]

$$= \frac{4+2}{4-2} = \frac{6}{2} = \boxed{\mathbf{3}}$$

Q8. Prove that: $(1 + \cot^2\theta)(1 - \cos\theta)(1 + \cos\theta) = 1$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Apply $1 + \cot^2\theta = \csc^2\theta$ and $(1 - \cos\theta)(1 + \cos\theta) = 1 - \cos^2\theta$ [1 Mark]

$$\text{LHS} = \csc^2\theta(1 - \cos^2\theta) = \csc^2\theta \cdot \sin^2\theta$$

Step 2: Simplify Reciprocals [1 Mark]

$$= \frac{1}{\sin^2\theta} \cdot \sin^2\theta = \boxed{\mathbf{1} = \text{RHS} \quad \text{[Hence Proved]}}$$

SECTION C: Short Answer Type — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. Prove that: $\dfrac{1}{\csc A - \cot A} - \dfrac{1}{\sin A} = \dfrac{1}{\sin A} - \dfrac{1}{\csc A + \cot A}$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Simplify LHS Using $\csc^2 A - \cot^2 A = 1$ [1.5 Marks]

$$\text{LHS} = \frac{\csc^2 A - \cot^2 A}{\csc A - \cot A} - \csc A = (\csc A + \cot A) - \csc A = \cot A$$

Step 2: Simplify RHS Analogously [1.5 Marks]

$$\text{RHS} = \csc A - \frac{\csc^2 A - \cot^2 A}{\csc A + \cot A} = \csc A - (\csc A - \cot A) = \cot A$$

$$\boxed{\therefore \text{LHS} = \text{RHS} = \cot \mathbf{A} \quad \text{[Hence Proved]}}$$

Q10. If $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$, show that $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Express $\sin\theta$ in Terms of $\cos\theta$ [1.5 Marks]

$$\cos\theta + \sin\theta = \sqrt{2}\cos\theta \implies \sin\theta = (\sqrt{2}-1)\cos\theta$$

Step 2: Multiply Both Sides by $(\sqrt{2}+1)$ [1.5 Marks]

$$(\sqrt{2}+1)\sin\theta = (\sqrt{2}+1)(\sqrt{2}-1)\cos\theta = (2-1)\cos\theta = \cos\theta$$

$$\sqrt{2}\sin\theta + \sin\theta = \cos\theta \implies \boxed{\cos\theta - \sin\theta = \sqrt{\mathbf{2}}\sin\theta \quad \text{[Hence Proved]}}$$

SECTION D: Long Answer / HOTS

MATHS MANTHAN | **(2 × 5 = 10 Marks)**

Q11. Prove that: $\dfrac{\tan^3\theta}{1+\tan^2\theta} + \dfrac{\cot^3\theta}{1+\cot^2\theta} = \sec\theta\csc\theta - 2\sin\theta\cos\theta$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Replace Denominators via $1+\tan^2\theta = \sec^2\theta$, $1+\cot^2\theta = \csc^2\theta$ [1.5 Marks]

$$\text{LHS} = \frac{\sin^3\theta/\cos^3\theta}{1/\cos^2\theta} + \frac{\cos^3\theta/\sin^3\theta}{1/\sin^2\theta} = \frac{\sin^3\theta}{\cos\theta} + \frac{\cos^3\theta}{\sin\theta}$$

Step 2: Combine Over $\sin\theta\cos\theta$ & Factor $\sin^4\theta + \cos^4\theta$ [2 Marks]

$$\text{LHS} = \frac{\sin^4\theta + \cos^4\theta}{\sin\theta\cos\theta} = \frac{(\sin^2\theta+\cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta}{\sin\theta\cos\theta} = \frac{1 - 2\sin^2\theta\cos^2\theta}{\sin\theta\cos\theta}$$

Step 3: Separate Terms to Obtain RHS [1.5 Marks]

$$\text{LHS} = \frac{1}{\sin\theta\cos\theta} - \frac{2\sin^2\theta\cos^2\theta}{\sin\theta\cos\theta} = \boxed{\sec\theta\csc\theta - \mathbf{2}\sin\theta\cos\theta = \text{RHS} \quad \text{[Hence Proved]}}$$

Q12. If $\dfrac{\cos\alpha}{\cos\beta} = m$ and $\dfrac{\cos\alpha}{\sin\beta} = n$, show that $(m^2+n^2)\cos^2\beta = n^2$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Substitute m and n into LHS [2 Marks]

$$\text{LHS} = (m^2+n^2)\cos^2\beta = \left(\frac{\cos^2\alpha}{\cos^2\beta} + \frac{\cos^2\alpha}{\sin^2\beta}\right)\cos^2\beta$$

Step 2: Take $\cos^2\alpha$ Common & Combine Fractions [2 Marks]

$$\text{LHS} = \cos^2\alpha\left(\frac{\sin^2\beta+\cos^2\beta}{\cos^2\beta\sin^2\beta}\right)\cos^2\beta = \cos^2\alpha\left(\frac{1}{\cos^2\beta\sin^2\beta}\right)\cos^2\beta$$

Step 3: Cancel $\cos^2\beta$ [1 Mark]

$$\text{LHS} = \frac{\cos^2\alpha}{\sin^2\beta} = \left(\frac{\cos\alpha}{\sin\beta}\right)^2 = \boxed{\mathbf{n^2} = \text{RHS} \quad \text{[Hence Proved]}}$$

Q13. Theodolite Geodetic Surveying of a Telecom Mast

A surveyor sets up a digital theodolite at station P on level ground to measure the height of a cellular communication tower TR. The telescope of the theodolite is mounted on a tripod at height $PQ = 1.5$ m above the ground. The surveyor measures the angle of elevation to the top of the tower T to be $\theta = 30°$. The horizontal distance between the tripod and the tower is $d = 30\sqrt{3}$ m. **Based on the above geometric model, answer the following questions:**

(i) Identify the right-angled triangle formed and write the relation for $\tan 30°$. [1]

(ii) Calculate the height of the tower section above the instrument level. [1]

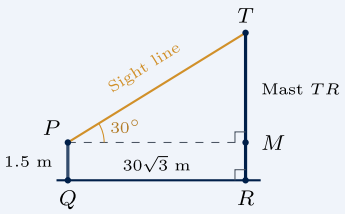
(iii) Calculate the total height of the cellular tower from ground level, and find the direct line-of-sight distance from the theodolite telescope to the tower top T. [2]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Sub-part (i): Right Triangle & $\tan 30°$ Relation [1 Mark]

In right-angled $\triangle PMT$ ($\angle M = 90°$, horizontal $PM = QR = 30\sqrt{3}$ m):

$$\boxed{\tan 30° = \frac{\textbf{TM}}{\textbf{PM}}}$$



Sub-part (ii): Mast Height Above Instrument Level (TM) [1 Mark]

$$TM = PM \tan 30° = 30\sqrt{3} \times \frac{1}{\sqrt{3}} \implies \boxed{TM = \textbf{30} \text{ m}}$$

Sub-part (iii): Total Mast Height TR & Direct Sight Line PT [2 Marks]

$$\text{Total Height } TR = TM + MR = 30 + 1.5 = \boxed{\textbf{31.5} \text{ m}}$$

$$PT = \frac{PM}{\cos 30°} = \frac{30\sqrt{3}}{\sqrt{3}/2} \implies \boxed{PT = \textbf{60} \text{ m}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★